

BPT-China Rules, Outline, & Syllabus

October 3, 2025

1 General Resources

Textbooks

- Daniel Kleppner and Robert Kolenkow—An Introduction to Mechanics, 2nd Edition [ISBN-13: 978-0521198110]
- Edward Murcell, David Morin—Electricity and Magnetism, 3rd Edition [ISBN-13: 978-1107014022]
- Douglas Giancoli—Physics: Principles with Applications (7th Edition) [ISBN-13: 978-0321625922]
- Richard Feynman—The Feynman Lectures on Physics [ISBN-13: 978-0201021165]

Practice Exams

- USAPhO and F=ma past exams
- Physics Brawl
- Chinese Physics Olympiad

2 Rules

There are *two* divisions within the competition: the lower division (pions) and the upper division (kaons). The main difference in difficulty is that the lower division will not include calculus-based problems, whereas the upper division may require the use of calculus while being generally more difficult.

Team competition will also be split into these divisions and competitors will remain in their division if they are assigned to a team. Teams consist of four students, and a team may be organized before or during the competition. Teams with three students and below may be sent extra students to ensure everyone is part of a team, but there is no inherent score disadvantage for not having four members.

3 Competitions Format

Saturday

- Part 1 - **Individual MCQ**
 - 30 Questions in 60 minutes
 - Generally easier questions on topics covered by this point in highschool level physics (up to AP equivalent) (E&M, Mechanics, wildcard problems that walk students through the process using math they should know).
- Part 2 - **Team Lab**
 - 1 lab, 60 minutes
 - Groups will receive sets of materials and a formula sheet, and will be tasked with something such as finding a constant or property of a material.
- Part 3 - **Team FRQ**
 - 9 Questions, each containing a few (3-4) subsections, totaling 90 minutes

- Will cover a wide variety of topics that stray from standard high school physics content to encourage teams to split up the questions according to their personal preferences.

Sunday

- Part 1 - **Individual FRQ**
 - 3 Questions, each containing 4-6 subsections, totaling 90 minutes with an easy, medium, and hard question.
- Part 2 - **GUTS**
 - 60 minutes
 - 5 sets of questions, each containing 3 parts.
 - Each correct part is worth an increasing number of points as the set number increases.
 - (e.g. Set 1: 10 points total; Set 2: 12 points; Set 3: 15 points ...)
 - Each team will choose one “runner” to retrieve the next set of problems after submitting the first set. No more than one set per team is allowed to be solved at a time.
 - We will compile a total of 7 sets in order of increasing difficulty. The easiest 5 will be used for the Pion division (junior) and the hardest 5 will be used for the Kaon division (senior).
- Part 3 - **Awards**
 - Thank helpers/participants/visitors, present awards for individuals and teams.

4 Scoring

4.1 Grading

Graders will use AtomicGrader to quickly organize and grade all multiple choice questions. FRQ style questions will be marked by a rubric by hand.

The penalty for incorrect answers on a multiple choice question is **1/4 point**, while a correct answer earns 1 point.

The rubric for the free response focuses on partial credit, so correct/final answers are not required to still score points. Questions are valued at a max of **between 1 and 4 points** based on certain thresholds and quantities of correct work.

The lab will be graded based on the procedure and the result, with a safe/accurate procedure scoring points and a result closer to the literature value scoring better.

GUTS will consist of **5 sets of questions** scored with x1 points, x1.1, x1.3, x1.6, and x2 point multipliers.

4.2 Point Allocation

The competition will be weighted as follows.

Individual

- 50% from multiple choice.
- 50% from free response.

Team

- 40% from free response.
- 40% from GUTS.
- 20% from the lab.

5 Syllabus

Pions:

- Math
 - Interpret, create, and analyze graphs of motion and forces.
 - Perform dimensional analysis.
 - Understanding of geometry, angles, and trigonometric functions.
- Kinematics/Mechanics
 - Draw and interpret free-body diagrams for objects in various contexts.
 - Projectile motion and range equation. Find the distance and height of a projectile in uniform gravity.
 - Apply Newton's laws of motion to solve dynamics problems.
 - Solve problems involving uniform and accelerated motion in one and two dimensions.
 - Basic application of Hooke's law. Sinusoidal motion of simple oscillators.
 - Analyze energy transformations and apply conservation of energy.
 - Use impulse-momentum theorem and conservation of momentum in elastic and inelastic collisions.
 - Apply simple rotational kinematics in 2 dimensions.
 - Basic understanding of torque, angular momentum, and rotational energy in 2 dimensions.
 - Relate linear and angular variables.
 - Solve equilibrium problems involving torques and forces.
 - Basic understanding of work and work-energy theorem.
- Fluid Dynamics
 - Pressure, flow of incompressible fluids.
 - Continuity of fluid motion.
 - Bernoulli's equation
- Electricity and Magnetism
 - Apply Coulomb's law and the principle of superposition for electric forces.
 - Use electric field and potential relationships, including equipotential diagrams.
 - Apply Ohm's law and Kirchhoff's rules to solve DC circuit problems.
 - Understand magnetic fields from currents and forces on charges in fields.
 - Analyze electromagnetic induction.
 - Apply ray diagrams, lens/mirror equations, and magnification.
 - Energy of systems of multiple charges.

Kaons:

- Calculus-Based Mechanics
 - Understand the calculus definitions of physical quantities.
 - Differentiate the position, velocity, and acceleration functions with respect to time.
 - Integrate acceleration or velocity functions to find velocity or position.
 - Solve motion problems with non-constant acceleration using differential equations.
 - Apply calculus to rotational motion (e.g. torque from distributed forces, moment of inertia via integration).
- Calculus-Based Electricity & Magnetism
 - Analyze capacitors, dielectrics, and RC circuits.
 - Calculate electric field and potential from continuous charge distributions using integrals.

- Solve problems using Gauss's law with symmetry arguments and calculus.
- Use Ampere's law to find magnetic fields from current distributions.
- Apply Faraday's law in integral form to calculate induced emf.
- Solve differential equations in RC, RL, and LC circuits for charge, current, or voltage as functions of time.
- Understand and apply Maxwell's equations.

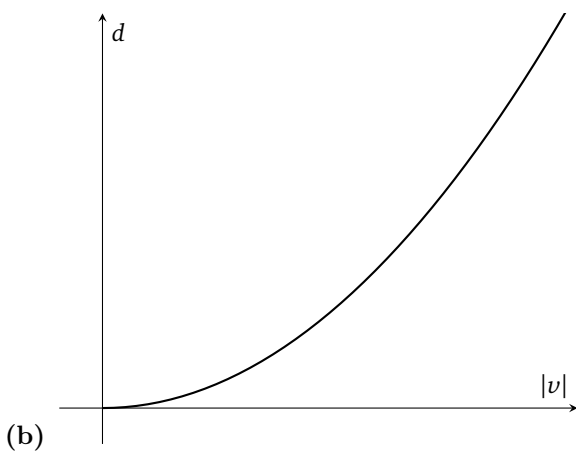
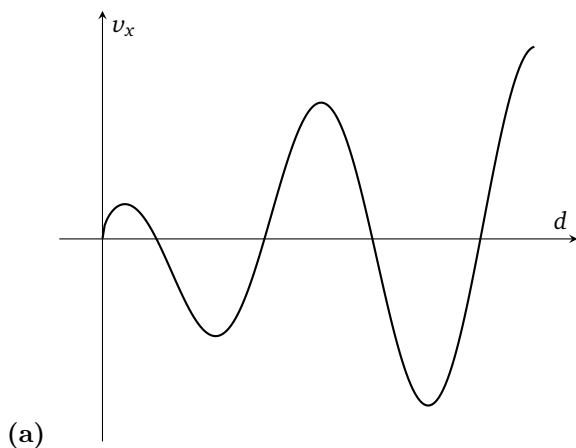
6 Sample Questions

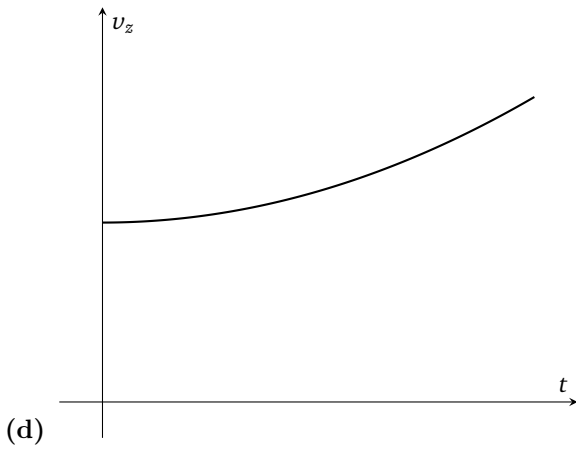
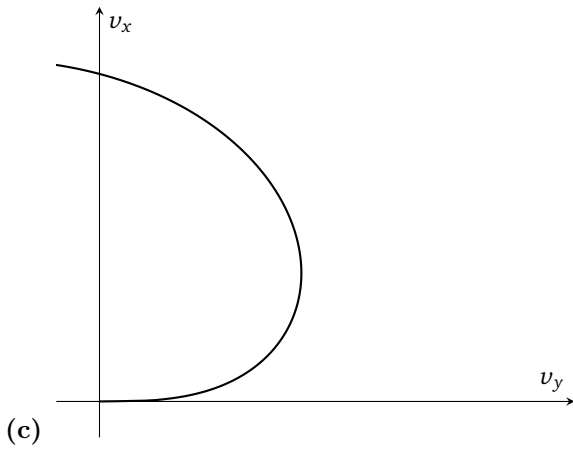
6.1 Individual MCQ

A ball rolls down a clockwise spiraling slide and experiences 2 revolutions around the slide while traveling downward. Assume no friction or air resistance. Recording begins when the ball lies at

$$(0, r, h)$$

where r is the radius of the slide and h is the height above the ground. Which of these graphs does NOT depict an accurate relationship in this system? v_x and v_y indicate horizontal velocities, v_z indicates vertical velocity, and d denotes the total distance traveled by the ball.





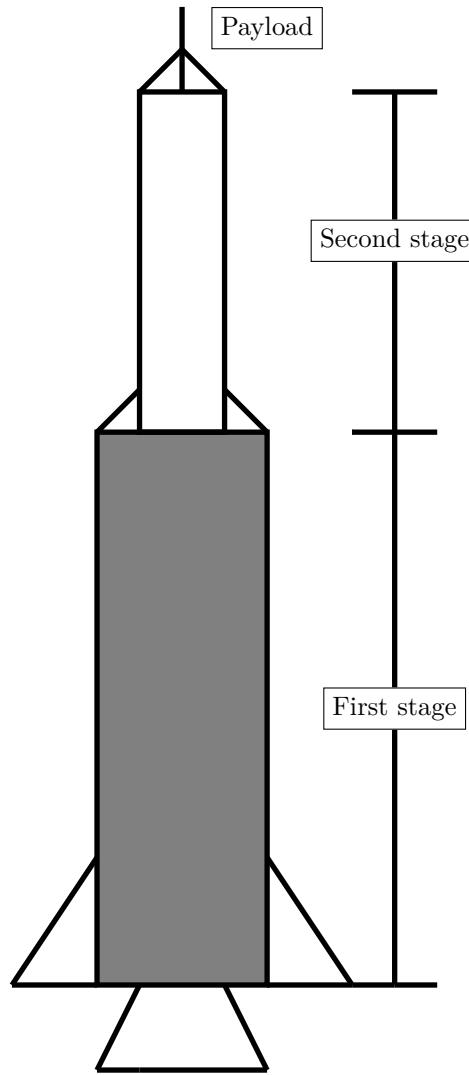
6.2 Team FRQ

Consider the following simplified two stage rocket. The rocket is delivering a payload, positioned at the top, with mass m . The second stage (directly under the payload) has a mass nm , and the total rocket, including the bottom stage, is Nm ($1 \leq n \leq N$). The fuel is ejected out of the engine at exit velocity of V .

You may use the following formula, commonly known as the rocket equation:

$$\vec{F}_{ext} = M \frac{d\vec{v}}{dt} - \vec{u} \frac{dM}{dt}$$

Where $\vec{F}_{ext}$ is the total external forces acting on the rocket, $\vec{v}$ is the rocket's velocity, M is the rocket's mass, and $\vec{u}$ is the exhaust velocity. Assume the stage separation is instant and exerts no force, and that the mass of the fuel tanks and engines are negligible compared to the mass of fuel.



- (a) Consider the two stage rocket in outer space with *no* external forces. What is the change in velocity Δv_1 after burning the first stage?
- (b) After the first stage is released, the second stage starts burning. Show that the final velocity after the second stage finishes burning does not depend on the value of n .
- (c) In a more realistic scenario, the rocket takes off from rest. Consider gravity and air resistance proportional to the velocity given by $\vec{F}_{fr} = -k\vec{v}$ and $\vec{F}_g = -M\vec{g}$. What is the maximum change in velocity in this case? Let the gravitational acceleration be g . You may assume the constant of proportionality of the air resistance and the gravitational acceleration does not change as the rocket gains altitude.
- (d) After exiting the atmosphere, the rocket drops the first stage and ignites the second stage. Now, the air is much thinner, so the air resistance is given by $\vec{F}_{fr} = -(k/2)\vec{v}$. At this altitude, you may assume gravity is negligible. What is the change in velocity achievable after the second burn?

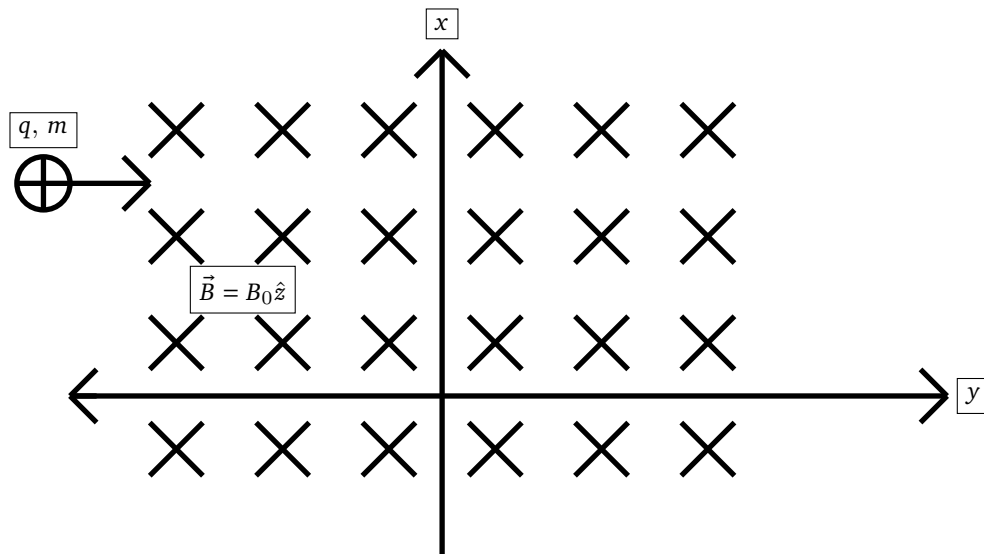
6.3 GUTS

A magician has a loaded hula hoop, which can be modeled as a uniform ring of mass m with a point mass of $m/3$ attached to a point on the ring. The magician places the ring on an incline with a very high friction coefficient, and finds the maximum angle θ in which the hoop can be set on the incline and remain stationary. What is θ ?

6.4 Individual FRQ

In plasma physics, we consider the collective behavior of charged particles in EM fields. To investigate

complicated problems, we usually first consider a simple version of it. Here we consider the motion of a single particle in EM fields.



- (a) With a uniform magnetic field $\vec{B} = B_0 \hat{z}$ in the space, if there is a charged particle with initial velocity $\vec{v} = v_0 \hat{y}$, charge q , and mass m , show that the charged particle will move in a circle, and find the frequency of that circular motion.
- (b) Now if we add another electric field $\vec{E} = E_0 \hat{y}$, how will the motion be different? Try to cancel out the Electric force by considering this in another reference frame.
- (c) Now let's consider the collective behavior: If there are many charged particles like this, with a number density of n (meaning that there are nV particles in a volume of V) show that the setup in (b) gives us a steady current, while (a) doesn't. This is actually a very simple model of Hall Effect, saying that if you put a piece of metal in $\vec{E}$ in y direction and $\vec{B}$ in z direction, there will be a current (or equivalently voltage) in x direction